

Table 3: Performance comparison of our proposed method with other baseline methods using BLEU on different datasets.

Method	MTD	CASICTB	CASIA2015	NEU
Base	18.21 (+0.00)	13.43 (+0.00)	18.65 (+0.00)	20.06 (+0.00)
Sentence Length	18.34 (+0.13)	13.52 (+0.09)	18.75 (+0.10)	20.21 (+0.15)
Word Rarity	18.35 (+0.14)	13.49 (+0.06)	18.72 (+0.07)	20.17 (+0.11)
Denoising	18.42 (+0.21)	13.55 (+0.12)	18.86 (+0.21)	20.44 (+0.38)
Ours-1 Round	18.81 (+0.60)	13.60 (+0.17)	18.91 (+0.26)	20.38 (+0.32)
Ours-2 Rounds	19.11 (+0.90)	14.03 (+0.60)	19.24 (+0.59)	20.79 (+0.73)

Table 4: Ablation analysis on different features of our proposed framework on MTD.

Method	MTD
Base	18.21 (+0.00)
Senlen + Logp	18.47 (+0.26)
+ N-gram	18.63 (+0.42)
+ Tagging	18.81 (+0.60)
+ 2nd Round	19.11 (+0.90)

Curriculum learning aims to organize the training process in a meaningful way by feeding certain samples to the model in certain training stage such that the model can learn better and faster (?). They propose a simple strategy which organizes all training samples into bins of similar complexity and starts training from the easiest bin to include more complexed bins until all bins are covered. (?) apply this idea to NMT by binning according to simple features like sentence length and word frequency, and improve this strategy by restricting that each sample can only be trained once during an epoch. (?) conduct empirical studies on several hand-crafted curriculum and adopt a probabilistic view of curriculum learning. (?) further propose a competence function $c(t)$ with respect to training time step t as the indicator of learning progress and select samples based on both difficulty and competence. However, these heuristic-based approaches highly depend on hand-crafted curriculum and are hard to generalize.

Compared with heuristic-based approaches, RL-based policy learning models are trained end-to-end and do not rely on hand-crafted strategies. (?) use Bayesian optimization to learn a linear model for ranking examples in a work-embedding task. (?) explore bandit optimization for scheduling tasks in a multi-task problem. (?) select examples in a co-trained classifier using RL. (?) organize the dataset into bins based on the data noise proposed by (?) and utilize a DQN to learn a data selection policy deciding from which bin to select the next batch.

Different from the existing curriculum learning methods, our work focuses on learning a training curriculum with reinforcement learning for an existing pre-trained NMT model. We argue that existing curriculum learning methods are only applicable on train-from-scratch scenarios, and learning from an existing model can save training time.

Active learning (?) is another related area which focuses on selectively obtaining labels for unlabeled data in order to improve the model with least labeling cost. (?) (?) study active learning for Statistical Machine Translation using some hard-coded heuristics. (?) design an active learning algorithm based on a deep Q-network, in which the action corresponds to binary annotation decisions applied to a stream of data. (?) make use of imitation learning to train a data selection policy.

6 CONCLUSION

In this paper, we study curriculum learning for NMT from a new perspective, to re-select a subset of useful samples from the existing dataset to further improve a pre-trained model, and formulate this task as a reinforcement learning problem. Compared with existing curriculum methods only applicable on train-from-scratch scenarios, our setting saves training time by better utilizing the existing pre-trained models. Our proposed framework is built based on the deterministic actor-critic algorithm, and learns a policy to select examples that can improve the model the most. We conduct experiments on several zh-en translation datasets and compare our method with other baseline methods including the easy-to-difficult curriculum and the denoising scheme. Through rounds of training and data selection, our method achieves a significant performance boost on the pre-trained model, and out-performs all baselines methods by a great margin.

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